

WASP-43b

R-1672

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220131 · created 2026-08-02T10:39:04+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459610.8833 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.015
Transit depth (Rp/Rs)^2	2.67 +/- 0.5 [%]
Orbital Inclination (inc)	83.57 +/- 1.39
Airmass coefficient 1 (a1)	1.409 +/- 0.015
Airmass coefficient 2 (a2)	-0.247 +/- 0.065
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	None
Optimal Aperture	3.64
Optimal Annulus	11.26
Transit Duration (day)	0.0547 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.015 vs 0.1594 (deviation 0.24σ)
Duration fit vs published	0.0547 d vs 0.0512 d (deviation 1.01σ)
Mid-time O–C	2.5 min
Depth SNR	10.9
Residual scatter	1.09 %

Light curve

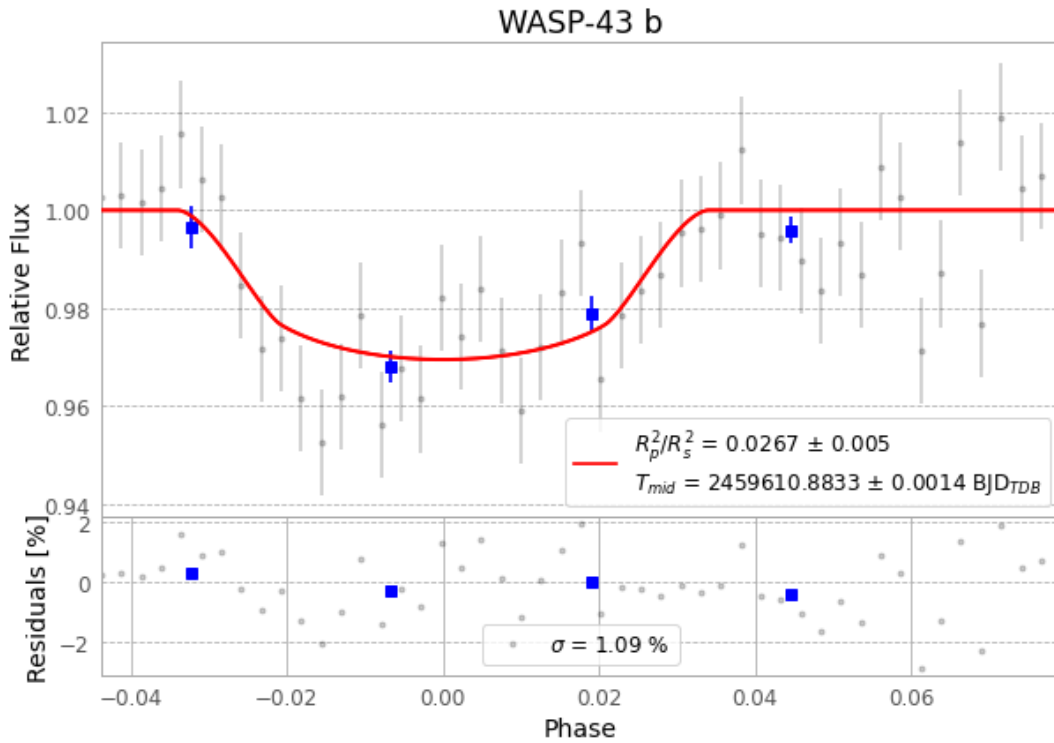


Figure 1 — FinalLightCurve_WASP-43 b_2022-01-31.png (SHA-256 e495e40d8bb38973...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 268], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 480d2d074e2edda6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

49 FITS frames (32 MB), WASP-43220131081212.FITS → WASP-43220131103613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220131.log (a87295d17f5c...), FinalLightCurve_WASP-43 b_2022-01-31.png (e495e40d8bb3...), FinalLightCurve_WASP-43 b_2022-01-31.csv (cc8d79d6526c...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 10:39Z.